LayerLogic

Real-Time Defect Detection in Advanced Additive Manufacturing via Multimodal Deep Learning

Teamname: SteelSync

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Track: Track A (Deep Tech for the World) & Track B (Sustainability)

Domain: Artificial Intelligence / Advanced Manufacturing

01 Problem Definition

Metal Additive Manufacturing (AM) encompassing processes like Laser Powder Bed Fusion (LPBF), Directed Energy Deposition (DED), and Wire Arc Additive Manufacturing (WAAM) is revolutionizing aerospace, defense, and biomedical engineering. It enables the fabrication of complex, topology-optimized, lightweight geometries that are mathematically impossible to create via traditional casting or subtractive CNC machining. However, the global adoption of metal AM is fundamentally bottlenecked by a severe and expensive failure mode: **stochastic, invisible micro-defects.**

The Core Physics Problem:

During printing, the highly concentrated energy source (laser or electron beam) creates a microscopic melt pool that is thermodynamically chaotic. Unpredictable thermal fluctuations, spatter ejection, vapor plume interference, and "keyhole-mode" collapses lead to the entrapment of shielding gases or the failure of metal powders to fuse properly. This results in sub-surface porosity, lack-of-fusion voids, and micro-cracking.

The Context & Waste Crisis:

Currently, metal AM operates largely in a "blind" open loop. Because these micro-defects form beneath the surface, they are invisible during the build. They are typically only discovered *after* the entire print is completed, utilizing highly expensive and time-consuming post-process X-ray Computed Tomography (CT) scans or destructive metallography.

If a severe defect occurs in layer 50 of a 5,000-layer print, the machine will obliviously continue printing for another 30 hours. **This reactive, post-mortem quality assurance regime results in massive, irreversible waste.** A single failed build consumes kilograms of highly expensive gas-atomized metal powder (e.g., Titanium or Inconel), wastes thousands of liters of argon shielding gas, and burns immense amounts of electrical energy. This directly contradicts the sustainability goals of Industry 4.0.

The Need:

To make metal AM sustainable and economically viable for serial production, there is a critical need for an in-process, purely software-driven monitoring system. We need a diagnostic engine capable of detecting sub-surface defects instantly as they form, **allowing the operator or machine controller to halt the failing build midway, thereby preventing the cascading waste of materials, energy, and machine time.**

[Insert Diagram: Melt Pool Thermodynamics and Keyhole Collapse]

02 Innovation Description

LayerLogic is a purely software-based, multimodal-AI "diagnostic brain" designed to monitor advanced metal 3D printing in real-time, layer-by-layer. Instead of modifying the heavy hardware of the printer, LayerLogic utilizes Edge AI to intelligently fuse distinct sensory data streams to evaluate structural integrity on the fly.

How it Works: Multimodal Sensor Fusion

LayerLogic reframes quality control from inspecting the physical object *afterward* to reading the physics of the process *as it happens*. It does this by fusing two independent, complementary data streams:

1. **Acoustic Emissions (The Sound):** High-frequency acoustic sensors (microphones/piezoelectrics) capture the acoustic signatures of the laser-matter interaction. Crucially, sound waves propagate from sub-surface events, capturing the acoustic signatures of phase changes, vapor-plume dynamics, and keyhole instability—events that are completely invisible to optical cameras. LayerLogic mathematically transforms these raw .wav time-series signals into **Mel-spectrograms** (time-frequency images).
2. **Optical Imagery (The Sight):** High-speed coaxial cameras capture the surface geometry, thermal gradients, and spatter ejection of the melt pool.

The AI Architecture:

LayerLogic utilizes lightweight Convolutional Neural Networks (CNNs, specifically optimized ResNet-18 variants suitable for edge deployment). The AI extracts deep mathematical feature vectors from *both* the acoustic spectrograms and the optical melt-pool images simultaneously. These features are concatenated and fused within a dense neural network layer to output a localized "Defect Probability Score" for the current print layer.

Core Value & Differentiation:

- **Waste Mitigation via Midway Halting:** By instantly flagging a critical lack-of-fusion event, LayerLogic triggers a system halt, saving all subsequent material and energy that would have been wasted on a doomed part.
- **Multimodal Cross-Validation:** Single-sensor optical systems suffer from high

false-positive rates because surface anomalies do not always equal structural defects. By forcing the AI to cross-validate what it "sees" with what it "hears," LayerLogic drastically reduces false alarms.

- **Process Agnostic:** While initially modeled on LPBF, the algorithmic fusion of sound and vision is easily retrainable for DED, WAAM, and even polymer extrusion, making it a universal Deep Tech framework.

[Insert CNN Sensor Fusion Architecture Diagram]

03 User Identification

LayerLogic is developed as a scalable deep-tech software solution targeting stakeholders who suffer the highest financial and material penalties from failed AM builds.

Stakeholder / Beneficiary	Primary Unmet Need	Impact of LayerLogic
Academic & National R&D Labs (e.g., IIT KGP, ISRO, DRDO)	Conducting highly experimental materials research where print failure rates are exceptionally high.	Prevents waste of limited grant funding on ruined metal powders and unnecessary CT scanning of failed test coupons.
AM Service Bureaus & Contract Manufacturers	Operating on tight profit margins. A single 40-hour print failure can wipe out a week's revenue and delay client delivery.	Enables "Qualify-As-You-Go" certification, drastically reducing scrap rates and improving machine utilization efficiency.
Aerospace & Medical Primes	Require stringent, layer-by-layer certification for safety-critical parts (e.g., turbine blades, orthopedic implants) to meet regulatory standards.	Provides a digital, cryptographic layer-wise health certificate, reducing the bottleneck of 100% post-process X-ray CT inspection.

[Insert Stakeholder Pain-Point Chart]

04 Existing Alternatives

Currently, the industry relies on outdated, reactive methods, or incomplete, single-sensor monitoring solutions to ensure part quality.

Existing Solution	How it Works	Why it is Insufficient	LayerLogic Advantage
Post-Process X-Ray CT & Destructive Testing	Scanning or cutting the part after the multi-day print is finished.	Highly reactive. The time, electricity, and expensive metal powder are already wasted by the time the defect is found.	**Proactive & Sustainable:** Detects flaws in milliseconds, allowing the print to be halted midway to save resources.
OEM Single-Sensor Monitors (e.g., Optical Tomography)	High-speed cameras provided by printer manufacturers that measure melt-pool radiation.	Blind to sub-surface thermodynamic events like trapped argon gas. High false-positive rates lead operators to ignore the alarms.	**Multimodal AI:** Cross-validates surface visual data with sub-surface acoustic data, providing deeply robust, physics-informed predictions.
Rule-Based Threshold Alerts	Software that stops the machine if sensor voltage spikes above a hard-coded limit.	Fails to account for the highly non-linear, transient nature of melt-pool dynamics, missing complex micro-defects.	**Deep Feature Extraction:** Uses CNNs to identify invisible mathematical patterns in data rather than relying on arbitrary human-set thresholds.

[Insert Competitor Comparison Table]

05 Technical Readiness

Current Technology Readiness Level (TRL): 3 (Analytical and Experimental Critical Function Proof of Concept)

As a student-led deep-tech initiative, LayerLogic is currently in the rigorous algorithmic prototyping phase. It is being developed as a highly optimized, local-compute pipeline to prove that expensive cloud architecture is not required for real-time inference.

Key Technical Architecture:

- **Data Processing Pipeline:** Python, librosa (for raw acoustic .wav to Mel-spectrogram transformation), and OpenCV (for optical image normalization and noise reduction).
- **Machine Learning Core:** PyTorch is utilized to build and train custom, lightweight ResNet-18 feature extractors and the subsequent fusion classification layers.
- **Edge Deployment:** The trained model is wrapped in a high-performance FastAPI application. This simulates real-world industrial deployment, proving that the software can accept streaming sensor data and return a defect prediction with millisecond latency on local hardware.

Major Technical Risks & Mitigation:

The primary risk in AI for manufacturing is the "Sim2Real" (Dataset-to-Machine) gap—a model trained on historical data may struggle with live sensor noise. To mitigate this, the immediate next milestone involves injecting synthetic Gaussian noise and simulating sensor drift during the training phase to ensure robust generalization before physical machine deployment.

[Insert Edge API Deployment Workflow]

06 Validation Status

To maintain rigorous scientific integrity for this competition, the validation status is strictly divided into completed algorithmic milestones on limited data and planned execution phases.

Completed Validation (Initial Subset Testing):

- **Architecture Design & Subset Validation:** Successfully mapped the theoretical multimodal CNN fusion architecture and conducted preliminary validation on a *limited subset* of open-source datasets (such as the NIST AM Material Database and specialized Kaggle AM acoustic/melt-pool datasets).
- **Physics-to-Math Translation:** Initial testing on this data subset confirmed the theoretical correlation between thermodynamic keyhole collapse and distinct acoustic/optical frequency anomalies, proving the core mathematical concept is sound.
- **Edge-Compute Viability:** Verified that the computational complexity of the feature extraction layers remains low enough for eventual edge-device (e.g., NVIDIA Jetson) deployment.

Planned Validation (Next Steps - Stage 2 Execution & Future Scope):

- **High-Fidelity Labeled Data Acquisition (Future Scope):** To transition from a proof-of-concept to a production-ready model, the critical next step is acquiring and integrating much larger, highly-labeled industrial datasets. Expanding beyond the initial subset is vital to refine model accuracy, eliminate false positives, and ensure robust generalization across different printer OEMs and metal alloys.
- **Full Model Training & Benchmarking:** Train the individual acoustic and optical

networks on the comprehensive datasets, followed by training the combined fusion architecture.

- **Quantitative Metrics:** Generate robust evaluation data.

  Placeholders for

  Insert Model Accuracy Graph

  Insert ROC Curve

  Insert Confusion Matrix

  will be populated to definitively prove the superiority of multimodal fusion over single-sensor detection.

- **Simulated Real-Time Inference:** Feed raw validation data through the local API dashboard in a streaming fashion to benchmark latency and prove that the software can trigger a "halting signal" faster than the deposition of the subsequent layer.

07 Intellectual Property Considerations

LayerLogic takes a software-first approach to advanced manufacturing, which dictates its IP strategy:

- **Academic IP & Patents:** The novel methodology of algorithmic sensor fusion specifically applied to real-time LPBF/DED defect classification presents patentable claims.
- **Trade Secrets (Weights & Architecture):** The true commercial value lies in the proprietary data preprocessing pipelines, the specific mathematical method of concatenating the multimodal feature vectors, and the resulting trained neural network weights.
- **Freedom to Operate:** The project is built entirely on open-source frameworks (PyTorch, Python, FastAPI) and utilizes open-access scientific datasets for initial training. This ensures clear freedom to operate and guarantees that the foundational software is not infringing on the closed-source, proprietary software of major 3D printer OEMs.

08 Commercialisation Pathway

While LayerLogic is currently an academic deep-tech project, it is architected with a clear pathway to real-world industrial deployment and scalability.

- **The Offering:** A deployable Deep Tech software package (Edge-inference API and localized monitoring dashboard).

- **Deployment Strategy:** Defense and aerospace companies absolutely refuse to send sensitive 3D printing data (e.g., missile turbine CAD models) to external cloud servers. LayerLogic is designed to be deployed on localized edge devices (like an industrial PC or NVIDIA Jetson) physically connected to the printer, ensuring 100% data privacy and bypassing corporate cloud security hurdles.
- **Scaling Model:**
  - *Phase 1 (Academic/Pilot):* Open-source the foundational architecture for the research community while deploying a pilot system at an IIT KGP manufacturing lab (e.g., CoE in Advanced Manufacturing).
  - *Phase 2 (B2B SaaS Licensing):* License the monitoring software to independent AM Service Bureaus as a third-party quality assurance upgrade for their legacy machines.
  - *Phase 3 (OEM Integration):* License the API directly to 3D printer manufacturers to embed as the native "smart brain" inside their next generation of hardware.

[Insert Deployment Pathway Diagram]

09 Implementation Roadmap

This roadmap outlines the plan to advance LayerLogic from its current algorithmic concept (TRL 3) to a lab-validated prototype (TRL 5) within a 12-month student-led framework.

Timeframe	**Key Milestones & Deliverables**
Months 0–2 (Current Phase)	**Data & Preprocessing:** Finalize extraction of NIST/Kaggle/Own datasets. Complete Python scripts for librosa acoustic-to-spectrogram conversion and OpenCV image normalization.
Months 3–5	**Core AI Training:** Train isolated optical and acoustic CNN models on a local machine. Implement feature concatenation and train the multimodal fusion model. Extract F1-scores and ROC curves.
Months 6–12	**Edge API Development:** Wrap the trained PyTorch model into a FastAPI application. Build a local graphical dashboard. Conduct simulated latency tests to ensure sub-second inference speeds. Partner with an institutional manufacturing lab. Feed live (or recorded) sensor data from an actual metal AM run into the

	LayerLogic API to validate real-world efficacy. Begin drafting provisional IP documents.
Future Vision	**Closed-Loop Control:** Transition from passive defect *detection* to active defect *prevention*. Instead of just halting the machine, the AI will feed parameter corrections (adjusting laser power/speed in milliseconds) back to the controller to "heal" the defect instantly.

[Insert Project Roadmap Timeline]

10 Funding Requirements

Total funding requested: ₹ 65,000

This proposal is designed as a foundational, open-source deep-tech contribution to the IIT Kharagpur advanced manufacturing ecosystem, rather than a commercial startup venture. Because model training and AI inference will utilize the institute's existing high-performance GPU computing clusters, this budget requires absolutely zero allocation for cloud computing or processing hardware.

The requested funding is minimized and strictly allocated to building a **Standardized Multimodal Data Rig** and funding the physical printing process. The core strategy is to fund the recording of highly controlled "benchmark prints" (e.g., an industrial gear geometry) on an existing campus AM machine. By capturing the acoustic and optical data of this specific print, LayerLogic will generate a high-fidelity dataset. This dataset and the foundational algorithm will then be open-sourced, allowing multiple future research groups at IIT KGP to continue the research, train new models, and iterate on the findings without needing continuous, expensive machine access.

| **Category** | **Estimated Allocation** | **Purpose & Justification** |

| **Standardized Multimodal Data Rig** | ₹ 40,000 | Procurement of a high-frequency piezoelectric acoustic sensor, a high-speed optical camera, data acquisition (DAQ) cables, and custom mounting hardware to safely instrument an existing campus 3D printer. |

| **Benchmark Print Consumables** | ₹ 25,000 | Material costs for executing the highly controlled benchmark prints. This covers the cost of the gas-atomized metal powder, argon shielding gas, and consumable substrate build plates. |

(Note: By leveraging campus computing resources and focusing solely on physical data

acquisition, this ultra-lean budget ensures 100% of the funds are directed toward creating a lasting, open-source research dataset for the institute).

[Insert Budget Allocation Pie Chart]

11 Closing Summary

- **Problem:** Metal Additive Manufacturing is severely bottlenecked by invisible, sub-surface defects. Continuing to print a flawed part for hours results in massive, unsustainable waste of expensive metal powders, shielding gas, and energy.
- **Innovation:** LayerLogic is a multimodal Deep Learning software pipeline that fuses acoustic and optical sensor data to detect complex thermodynamic defects in real-time, allowing operators to halt failing prints midway.
- **Evidence:** The core architecture is firmly grounded in state-of-the-art manufacturing physics and relies on proven open-source industrial datasets (NIST), circumventing the need for immediate, expensive hardware access.
- **Next Milestone:** With the requested competition support, LayerLogic will transition from a local laptop algorithm into an optimized Edge-AI pilot, proving that sustainable, zero-waste Industry 4.0 requires intelligent software, not just heavy machinery.
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